(i)

$$a(x+y+z) - (ax+by+cz) = (a-b)y + (a-c)z$$

$\because a > b$, $a > c$ and $y, z \geq 0$

$$\therefore \boxed{ax + by + cz \leq a(x+y+z)}$$

(ii)(a) Δ is the sum of the areas of triangles PBC, PCA, PAB, with bases a, b, c and heights x, y, z

$$\therefore \boxed{\Delta = \tfrac{1}{2}(ax + by + cz)}$$

(ii)(b) By part (i),

$$2\Delta = ax + by + cz \leq a(x+y+z) \implies x + y + z \geq \frac{2\Delta}{a}$$

Minimum occurs when $y = z = 0$, giving $ax = 2\Delta$

$$\therefore \boxed{\min(x+y+z) = \frac{2\Delta}{a}, \quad \text{at } x = \frac{2\Delta}{a},\ y = z = 0}$$

(iii)(a)

$$\begin{aligned}(bx-ay)^2 + (cy-bz)^2 + (az-cx)^2 + (ax+by+cz)^2 = {} & b^2x^2 - 2abxy + a^2y^2 \\ & + c^2y^2 - 2bcyz + b^2z^2 \\ & + a^2z^2 - 2acxz + c^2x^2 \\ & + a^2x^2 + b^2y^2 + c^2z^2 \\ & + 2abxy + 2bcyz + 2acxz\end{aligned}$$

$$= \left(a^2+b^2+c^2\right)\left(x^2+y^2+z^2\right)$$

(iii)(b) The first three squares are non-negative, so

$$\left(a^2+b^2+c^2\right)\left(x^2+y^2+z^2\right) \geq (ax+by+cz)^2 = 4\Delta^2$$

$$x^2 + y^2 + z^2 \geq \frac{4\Delta^2}{a^2+b^2+c^2}$$

Minimum occurs when $bx = ay$, $cy = bz$, $az = cx$, i.e.

$$\frac{x}{a} = \frac{y}{b} = \frac{z}{c} = k, \qquad k\left(a^2 + b^2 + c^2\right) = 2\Delta$$

$$\therefore \boxed{\min\left(x^2 + y^2 + z^2\right) = \frac{4\Delta^2}{a^2 + b^2 + c^2}, \text{ at } x = \frac{2a\Delta}{a^2 + b^2 + c^2},\ y = \frac{2b\Delta}{a^2 + b^2 + c^2},\ z = \frac{2c\Delta}{a^2 + b^2 + c^2}}$$

(iv) c is the smallest of a, b, c, so as in part (i)

$$c(x + y + z) \leq ax + by + cz = 2\Delta \implies x + y + z \leq \frac{2\Delta}{c}$$

$x, y, z \geq 0$, so

$$x^2 + y^2 + z^2 \leq (x + y + z)^2 \leq \frac{4\Delta^2}{c^2}$$

Maximum occurs when $x = y = 0$, giving $cz = 2\Delta$

$$\therefore \boxed{\max\left(x^2 + y^2 + z^2\right) = \frac{4\Delta^2}{c^2}, \text{ at } x = y = 0,\ z = \frac{2\Delta}{c}}$$

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